David GILLIAM **MATH 1351 Spring 2004**

Homework Set HW11 due 4/1/04 at 11:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)
This is problem 2 Section 4.7 page 259.

Given the cost of producing x units of a product $C(x) = (x+30)/5$ and the selling price $p(x) = \dfrac{(70-x)}{(x+30)}$, determine the level of production x that minimizes profit.

$x =$ ______

2.(1 pt)
This is problem 4 Section 4.7 page 259.

Suppose the total cost of producing x units of a certain commodity is

$$C(x) = 2x^4 - 10x^3 - 18x^2 + x + 5 \text{ dollars}$$

Determine the largest and smallest values of the marginal cost for $0 \leq x \leq 5$.

largest value of marginal cost: ______

smallest value of marginal cost: ______

3.(1 pt)
This is problem 5 Section 4.7 page 259.

Suppose the total cost of manufacuring x units of a certain commodity is

$$C(x) = 3x^2 + x + 48 \text{ dollars}$$

Determine the minimum average cost.

minimum average cost: ______

4.(1 pt)
This is problem 6 Section 4.7 page 259.

A toy manufacturer produces an inexpensive doll (Flopsy) and an expensive doll (Mopsy) in units of x hundreds and y hundreds, respectively. Suppose it is possible to produce the dolls in such a way that

$$y = \frac{82 - 10x}{10 - x}, \quad 0 \leq x \leq 8$$

and that the company receives *twice* as much for selling a Mopsy doll as for selling a Flopsy doll. Find the level of production for both x and y for which the total revenue derived from selling these dolls is maximized. (NOTE: here we make the assumption that the revenue function is continuous)

Number of Flopsy dolls: $x =$ ______

Number of Mopsy dolls: $y =$ ______

5.(1 pt)
This is problem 8 Section 4.7 page 259.

Suppose a manufacturer estimates that, when the market price of a certain product is p, the number of units sold will be

$$x = -6\ln\left(\frac{p}{40}\right)$$

It is also estimated that the cost of producing these x units will be

$$C(x) = 4xe^{-x/6} + 30$$

a) Find the average cost, the marginal cost, and the marginal revenue for this production process.

Average Cost: ______

Marginal Cost: ______

Marginal Revenue: ______

b) What level of production x corresponds to maximum profit?

$x =$ ______

6.(1 pt)
This is problem 10 Section 4.7 page 259.

A certain industrial machine depreciates in such a way that its value (in dollars) after t years is

$$Q(t) = 20000e^{-.4t}$$

a) At what rate is the value of the machine changing with respect to time after 5 years?

Rate: ______

b) At what percentage rate is the value of the machine changing with respect to time after t years ?

Percentage Rate: ______ *percent*

7.(1 pt)
This is problem 12 Section 4.7 page 259.

Some psychcologists model a child's ability to memorize by a function of the form

$$g(t) = \begin{cases} t\ln(t) + 1 & \text{if } 0 < t \leq 4 \\ 1 & \text{if } t = 0 \end{cases}$$

where t is time, measured in years. Determine when the largest and smallest years of g occur.

Largest: $t =$ ______

Smallest: $t =$ ______

8.(1 pt)
This is problem 16 Section 4.7 page 259.

Suppose the total cost (in dollars) of manufacturing x units of a certain commodity is

$$C(x) = 3x^2 + 5x + 75$$

a) At what level of production x is the average cost per unit the smallest? ______

b) At what level of production x is the average cost per unit equal to the marginal cost? $x =$ ______

9.(1 pt)
This is problem 20 Section 4.7 page 260.

Suppose you own a parcel of land whose value t years from now is modeled as

$$P(t) = 200\ln(\sqrt{2t}) \text{ (in thousands of dollars)}$$

If the prevailing interest rate is 10 percent compounded continuously, when will be the most advantageous time to sell?

$t =$ ______

NOTE: leave your answer as a decimal, e.g. $t = 3.724$.

10.(1 pt)
This is problem 24 Section 4.7 page 260.

A bookstore can obtain the best-seller *20,000 Leagues Under the Majors* from the publisher at a cost of \$ 6 per book. The store has been offering the book at the price of \$ 30 per copy and has been selling 200 copies per month at this price. The bookstore is planning to lower its price to stimulate sales and estiamtes that for each \$ 2 reduction in price, 20 more books will be sold per month. At what price should the bookstore sell the sell the book to generate the greatest profit.

Price: \$ ______

11.(1 pt)
This is problem 28 Section 4.7 page 260.

A commuter train carries 600 passengers each day from a suburb to a city. It now costs \$ 5 per person to ride the train. A study shows that 50 additional people will ride the train for each 25 cents reduction in fare. What fare should be charged to maximize total revenue?

Price: \$ ______

12.(1 pt)
This is problem 32 Section 4.7 page 261.

Homing pigeons will rarely fly over large bodies of water unless forced to do so, presumably because it requires more energy to maintain altitude in flight over the cool water. Suppose a pigeon is released from a boat floating on a lake 3 miles from a point A on the shore and 10 miles away from the pigeon's loft (see Figure 4.84, page 261).

Assuming the pigeon requires twice as much energy to fly over water as over land and it follows a path that minimizes total energy, find the angle θ of its heading as it leaves the boat.

$\theta =$ ______ *radians*

13.(1 pt)
This is problem 4 Section 5.1 page 281.

Find the indefinite integral

$\int (4-5x)\,dx =$ ______ $+C$

14.(1 pt)
This is problem 6 Section 5.1 page 281.

Find the indefinite integral

$\int (-8t^3+15t^5)\,dt =$ ______ $+C$

15.(1 pt)
This is problem 8 Section 5.1 page 281.

Find the indefinite integral

$\int 14e^x\,dx =$ ______ $+C$

16.(1 pt)
This is problem 12 Section 5.1 page 281.

Find the indefinite integral

$\int \sec(t)\tan(t)\,dt =$ ______ $+C$

17.(1 pt)
This is problem 14 Section 5.1 page 281.

Find the indefinite integral

$\int \frac{\cos(t)}{3}\,dt =$ ______ $+C$

18.(1 pt)
This is problem 16 Section 5.1 page 281.

Find the indefinite integral

$\int \frac{dx}{10(1+x^2)} =$ ______ $+C$

19.(1 pt)
This is problem 18 Section 5.1 page 281.

Find the indefinite integral

$\int (x^3-3x+\sqrt[4]{x}-5)\,dx =$ ______ $+C$

20.(1 pt)
This is problem 22 Section 5.1 page 281.

Find the indefinite integral

$\int \frac{1}{t}\left(\frac{2}{t^2}-\frac{3}{t^3}\right)dt =$ ______ $+C$

21.(1 pt)
This is problem 24 Section 5.1 page 281.

Find the indefinite integral

$\int (3-4x^3)^2\,dx =$ ______ $+C$

22.(1 pt)
This is problem 26 Section 5.1 page 281.

Find the indefinite integral

$\int \frac{x^2+\sqrt{x}+1}{x^2}\,dx =$ ______ $+C$